

Conducting Forensic Investigations on System Memory (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 10

Student:

Bibek Ghimire

Email:

ghimirebibek0227@gmail.com

Time on Task:

2 hours, 38 minutes

Progress:

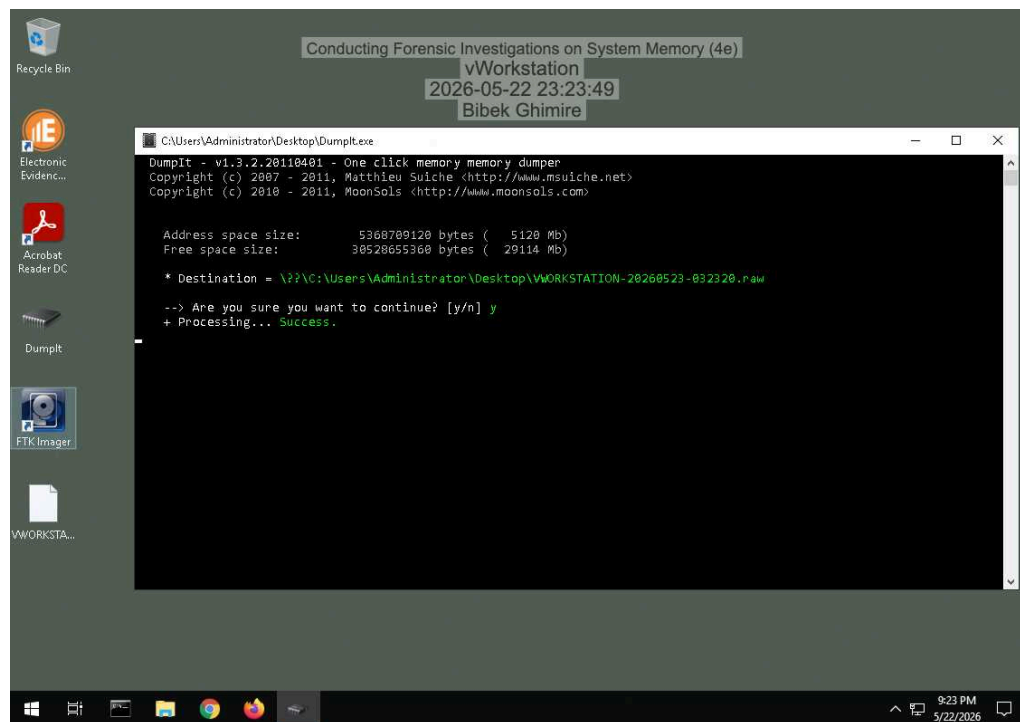
100%

Report Generated: Thursday, June 25, 2026 at 4:45 PM

Section 1: Hands-On Demonstration

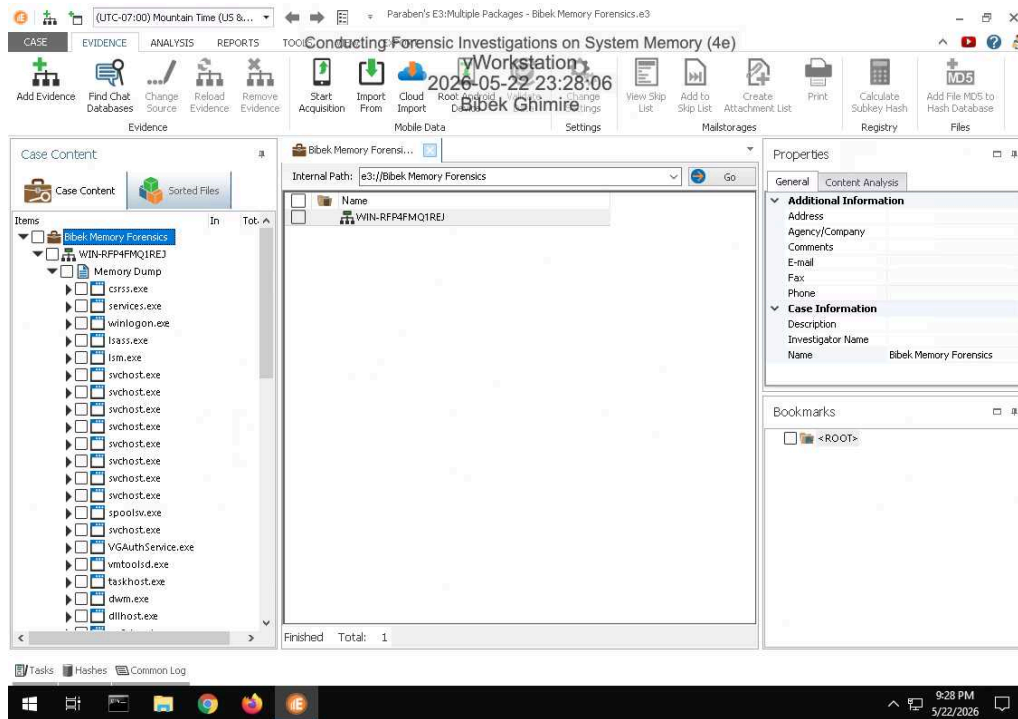
Part 1: Capture Memory using DumpIt

3. Make a screen capture showing the **Dumplt** success notification.



Part 2: Analyze Memory using E3

8. Make a screen capture showing the list of processes in the memory dump.



10. Record the start times for the oldest process and the newest process.

The start time for the oldest process is 7/12/2021 4:24:49 AM and the newest processes start time was 7/12/2021 6:42:43 AM.

15. Document your findings for the conhost.exe process. What is it and what is it used for?

The Conhost.exe process is used for core services to the Command Prompt in which it is used for feature's that allow Command Prompt to work with Windows Explorer to drag and drop files and folders to the terminal window. It stands for "Console Window Host" and is a legitimate program but can be susceptible to vulnerabilities caused by hacker's.

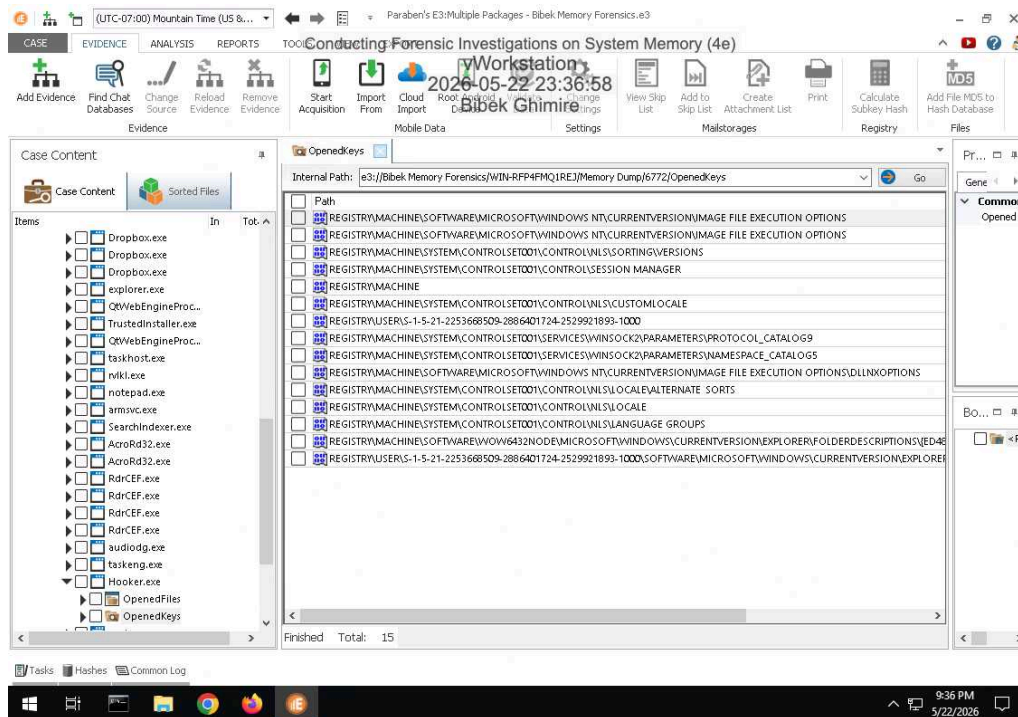
17. Document your findings for the hooker.exe process. What is it and what is it used for?

The Hooker.exe process is not a Windows system file and is used to record keyboard shortcuts as well as connect to the internet, record keyboard and mouse inputs and monitor applications. It is not considered a secure application and is vulnerable to attacks.

Conducting Forensic Investigations on System Memory (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 10

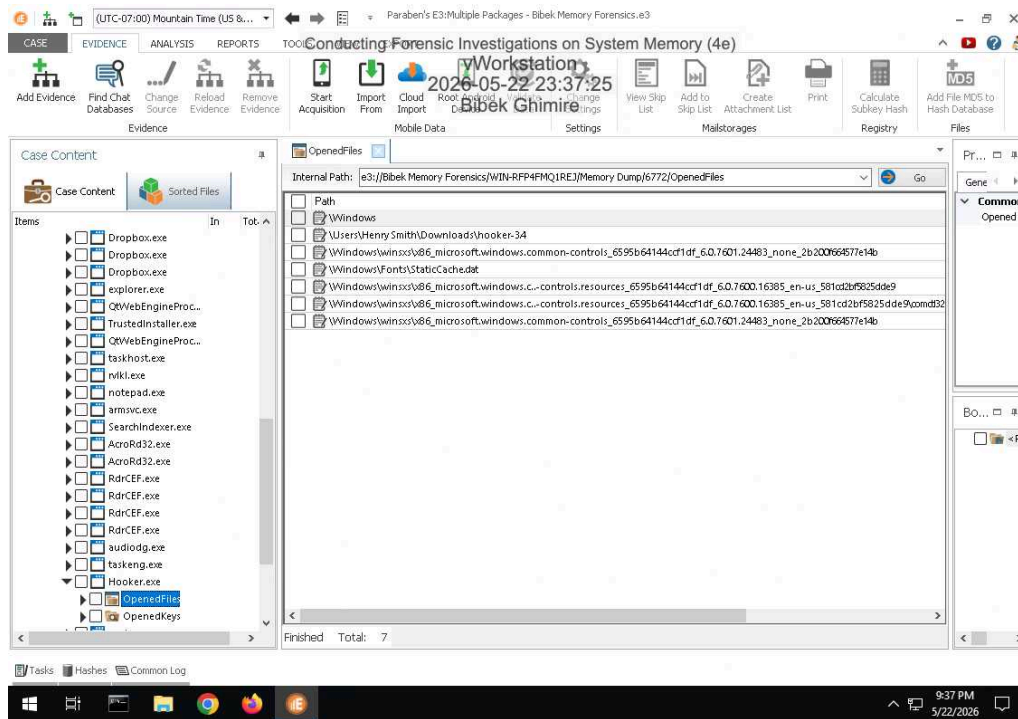
21. Make a screen capture showing the registry keys opened by the Hooker.exe process.



Conducting Forensic Investigations on System Memory (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 10

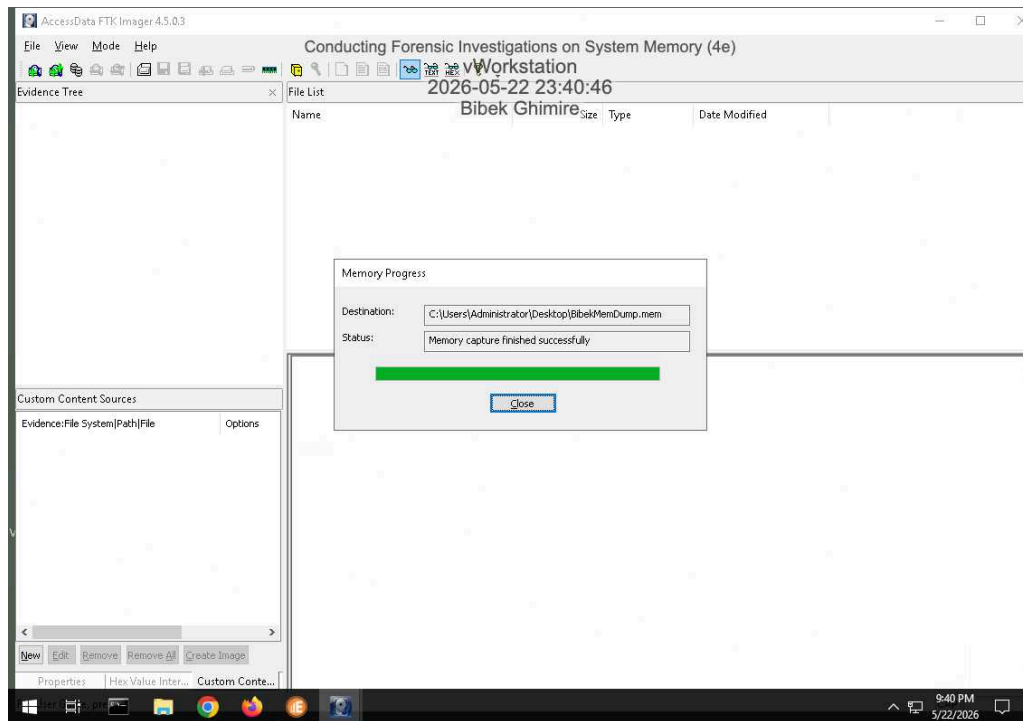
23. Make a screen capture showing the files opened by the hooker.exe process.



Section 2: Applied Learning

Part 1: Capture Memory using FTK Imager

6. Make a screen capture showing the *Memory capture finished successfully* confirmation.



Part 2: Analyze Memory using Volatility

7. **Document** your findings for the rvkl.exe process. What is it and what is it used for?

The rvkl.exe process is a security and monitoring software tool which is named "Logixoft's Revealer Keylogger." It creates log files of all usage that includes screenshots and logs that can be sent to the customer remotely. This program is not visible and has a digital signature and is able to record keyboard and mouse inputs and monitor inputs. It is not considered fully safe to use and may not work properly with some layouts.

9. **Document** whether any processes are flagged as hidden.

There are three different processes that are flagged as hidden: 1. services.exe 2. lsass.exe 3. lsm.exe.

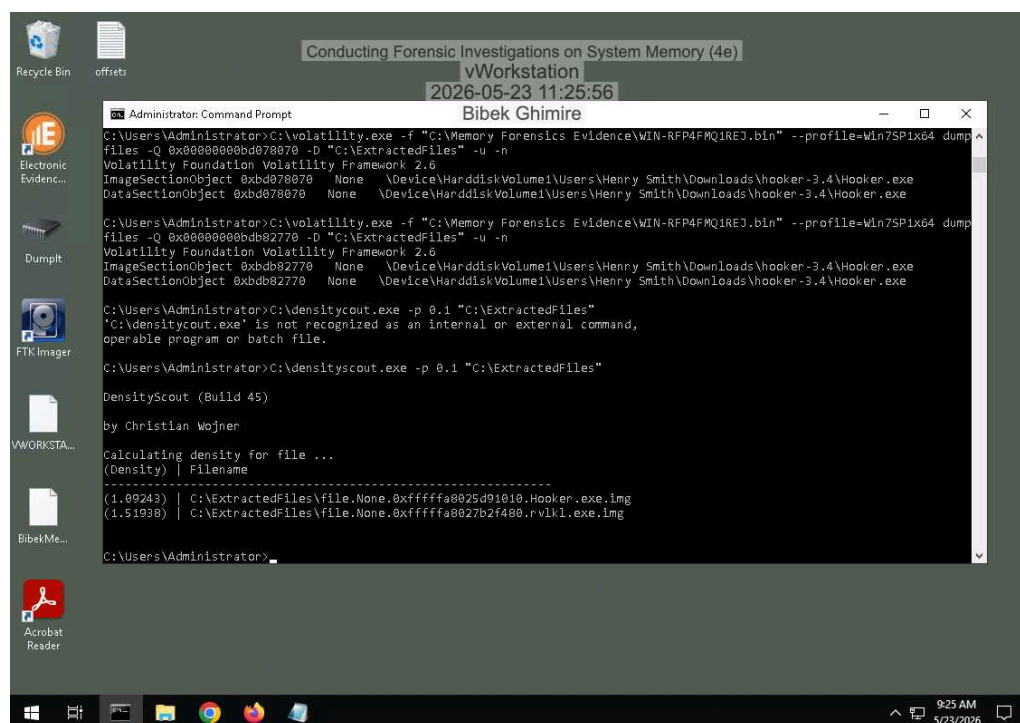
12. **Document** whether the netscan module displays network usage associated with the Hooker.exe or rvlkl.exe processes.

Yes, the netscan module displays that there is network usage associated with the Hooker.exe or rvlkl.exe processes through the Chrome Browser and Dropbox applications.

15. **Document** any information you were able to gather about port 56610.

The information I got about port 56610 was that it is a dynamic and private port that is used for accessing files on the system. The program name for the filesystem access tools is Xsan or Xsan Filesystem Access.

26. **Make a screen capture** showing the DensityScout results.



Section 3: Challenge and Analysis

Part 1: Identify Malicious Connections

Document the three processes that connected to 205.134.253.10:4444.

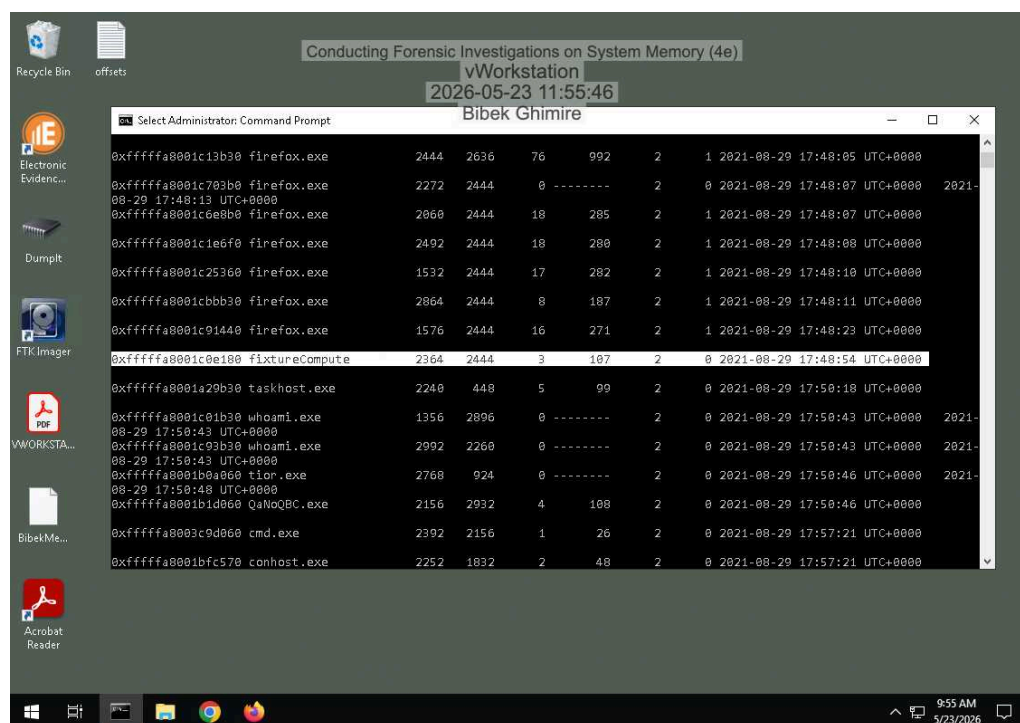
The three processes that connected to 205.134.253.10:4444 are "fixtureCompute," "QaNoQBC.exe," and "dllhost.exe."

Document the name and purpose of the software you discovered.

The software that commonly uses port 4444 as its default listener is Metasploit and also a proxy named 12P HTTP/S. Metasploit is a security software programming tool and project used to provide data about vulnerabilities and assists in penetration testing.

Part 2: Identify Malicious Processes

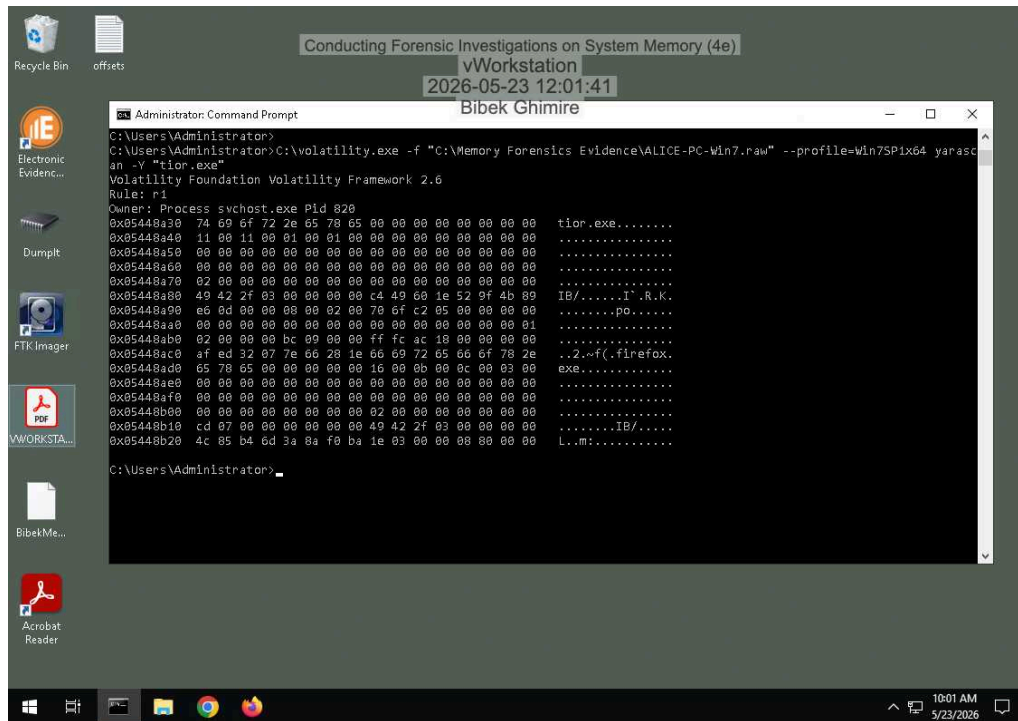
Make a screen capture showing the `fixtureComputer.exe` process, and all those below it, in the `pslist` output.



Conducting Forensic Investigations on System Memory (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 10

Make a screen capture showing the output of the yarascan.

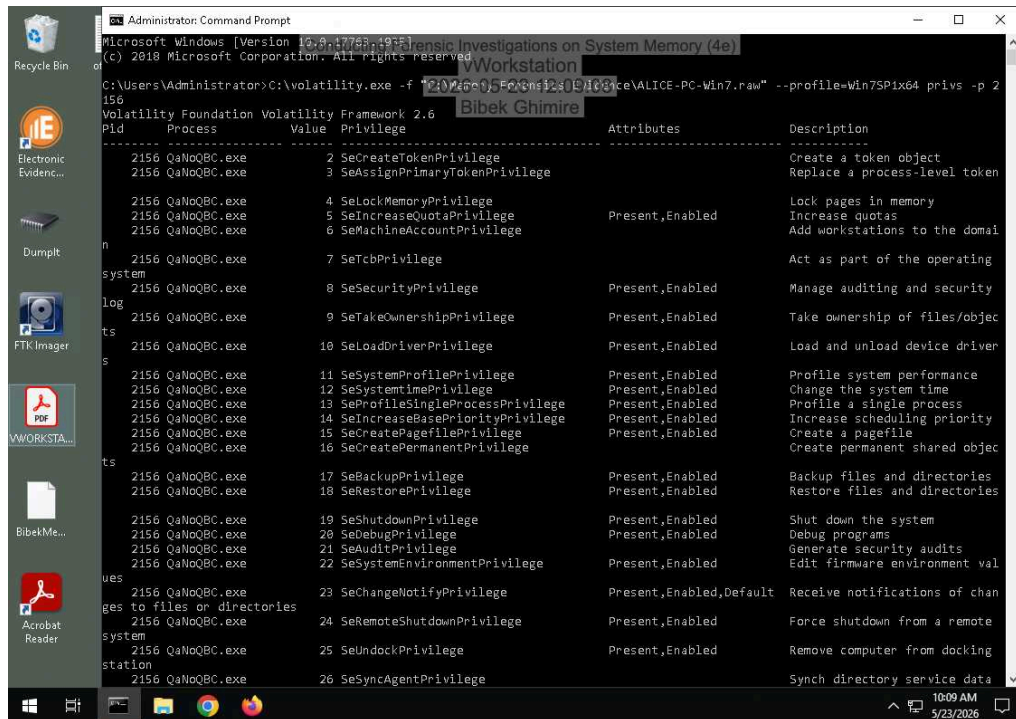


Part 3: Identify Privilege Escalation

Conducting Forensic Investigations on System Memory (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 10

Make a screen capture showing the output of your privilege comparison.



Administrator: Command Prompt

Microsoft Windows [Version 10.0.17763.195]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>C:\volatility.exe -f "C:\Users\Administrator\Documents\Bibek Chimire\Workstation\QaNoqBC.exe\ALICE-PC-Win7.raw" --profile=Win7SP1x64 privs -p 2156

Volatility Foundation Volatility Framework 2.6

Pid	Process	Value	Privilege	Attributes	Description
2156	QaNoqBC.exe	2	SeCreateTokenPrivilege		Create a token object
2156	QaNoqBC.exe	3	SeAssignPrimaryTokenPrivilege		Replace a process-level token
2156	QaNoqBC.exe	4	SeLockMemoryPrivilege	Present,Enabled	Lock pages in memory
2156	QaNoqBC.exe	5	SeIncreaseQuotaPrivilege		Increase quotas
2156	QaNoqBC.exe	6	SeMachineAccountPrivilege		Add workstations to the domain
2156	QaNoqBC.exe	7	SeTcbPrivilege		Act as part of the operating system
2156	QaNoqBC.exe	8	SeSecurityPrivilege	Present,Enabled	Manage auditing and security log
2156	QaNoqBC.exe	9	SeTakeOwnershipPrivilege	Present,Enabled	Take ownership of files/objects
2156	QaNoqBC.exe	10	SeLoadDriverPrivilege	Present,Enabled	Load and unload device drivers
2156	QaNoqBC.exe	11	SeSystemProfilePrivilege	Present,Enabled	Profile system performance
2156	QaNoqBC.exe	12	SeSystemTimePrivilege	Present,Enabled	Change the system time
2156	QaNoqBC.exe	13	SeProfileSingleProcessPrivilege	Present,Enabled	Profile a single process
2156	QaNoqBC.exe	14	SeIncreaseBasePriorityPrivilege	Present,Enabled	Increase scheduling priority
2156	QaNoqBC.exe	15	SeCreatePagefilePrivilege	Present,Enabled	Create a pagefile
2156	QaNoqBC.exe	16	SeCreatePermanentPrivilege		Create permanent shared objects
2156	QaNoqBC.exe	17	SeBackupPrivilege	Present,Enabled	Backup files and directories
2156	QaNoqBC.exe	18	SeRestorePrivilege	Present,Enabled	Restore files and directories
2156	QaNoqBC.exe	19	SeShutdownPrivilege	Present,Enabled	Shut down the system
2156	QaNoqBC.exe	20	SeDebugPrivilege	Present,Enabled	Debug programs
2156	QaNoqBC.exe	21	SeAuditPrivilege		Generate security audits
2156	QaNoqBC.exe	22	SeSystemEnvironmentPrivilege	Present,Enabled	Edit firmware environment variables
2156	QaNoqBC.exe	23	SeChangeNotifyPrivilege	Present,Enabled,Default	Receive notifications of changes to files or directories
2156	QaNoqBC.exe	24	SeRemoteShutdownPrivilege	Present,Enabled	Force shutdown from a remote system
2156	QaNoqBC.exe	25	SeUndockPrivilege	Present,Enabled	Remove computer from docking station
2156	QaNoqBC.exe	26	SeSyncAgentPrivilege		Synch directory service data

10:09 AM 5/23/2026